

UNIT 3

PROGRAMMING

FUNDAMENTALS

10TH COMPUTER SCIENCE FEDERAL BOARD

AWINASH GOSWAMI

AGENDA

- Front-end
- Back-end



BASIC TERMINOLOGIES CONT....

- **Front-End Development:** It is the part of web development that deals with everything the user sees and interacts with directly in their browser. It focuses on the look, feel, and behavior of a website or web application.
- **Key Technologies:**
 - *HTML – Structure (headings, paragraphs, images)*
 - *CSS – Styling (colors, fonts, layout)*
 - *JavaScript – Interactivity (buttons, animations, forms)*
- **Examples of what front-end developers do:**
 - *Design menus, buttons, and forms*
 - *Make a website look good on mobile and desktop*
 - *Create smooth animations and pop-ups*
 - *Ensure fast loading and easy navigation*



Markup Language
Content
HTML



Style sheet Language
Presentation
CSS



Programming Language
Behavior
JavaScript

BASIC TERMINOLOGIES CONT....

- **Back-End Development:** It is the part of web development that deals with everything behind the scenes – the server, database, and application logic that users do not see. It makes sure the website works properly, stores data, and processes user requests.
- **Key Technologies:**
 - *Languages: Python, PHP, Java, Ruby, C#, JavaScript (Node.js)*
 - *Databases: MySQL, PostgreSQL, MongoDB (to store user data)*
 - *Servers: Apache, Nginx (to host and run the website)*
- **Examples of what back-end developers do:**
 - *Store user login details in a database*
 - *Process online payments securely*
 - *Send emails or notifications*
 - *Fetch and send data to the front-end when a user clicks a button*

